Phase 1: NYC TLC Yellow Taxi, January 2024

1. Problem and decision

The operational question is whether recent, available trip history can improve next-hour pickup demand forecasts by TLC taxi zone, supporting vehicle allocation. Unit of source data: one provider-submitted trip row. Prediction unit: zone-hour, including future zero-count cells. This observational reporting source covers Yellow Taxi trips, not all NYC mobility.

2. Source and study population

Official monthly Parquet: https://d37ci6vzurychx.cloudfront.net/trip-data/yellow_tripdata_2024-01.parquet (49,961,641 bytes). TLC documentation: https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page ; dictionary: https://www.nyc.gov/assets/tlc/downloads/pdf/data_dictionary_trip_records_yellow.pdf . TLC receives vendor submissions and does not guarantee their accuracy or completeness.

Table 1. Source and analysis populations.

Population	Rows
Raw provider rows	2,964,624
Pickup time in [2024-01-01, 2024-02-01)	2,964,606
In month and pickup zone ID 1-263	2,952,588
Previous trip-quality filtered set, comparison only	2,913,207

There are 18 out-of-month pickups (15 before January; 3 in/after February). Minimum timestamp is 2002-12-31 22:59:39; it is an anomalous record, not evidence of a 2002 study period. An additional 12,018 in-month rows have pickup zone IDs outside the selected 1-263 study population. Codes 264 (Unknown/N/A) and 265 (Outside of NYC) are recognized special TLC codes, not automatically corrupt identifiers. Counts of demand use all other in-month pickup events regardless of fare, trip duration, or derived speed. No raw rows are deleted. Join the official taxi-zone lookup using PULocationID or DOLocationID against LocationID; the study range also includes Newark Airport and is not exclusively the five NYC boroughs. Membership of every observed code in the official lookup has not been independently checked.

3. Structure, quality and cleaning decisions

The file has 19 source columns. The full column names, observed dtypes and per-column missing counts/percentages are in outputs/schema_missing.csv. VendorID is a provider code; PULocationID/DOLocationID are zone keys, not unique trip IDs. There is no verified unique trip identifier. Exact duplicates compare every original source column: 0; shared time/zone alone is not a duplicate.

Table 2. Quality flags across raw records.

Rule (denominator: raw rows)	Count	Interpretation
Missing passenger_count	140,162	Missing, retain
Passenger zero / negative / >6	31,465 / 0 / 60	Zero/suspicious; inspect, retain for demand
Nonpositive duration / reversed time	870 / 56	Impossible elapsed trip for duration analysis
Negative / zero distance	0 / 60,371	Zero may be special trip
Negative fare / nonpositive total	37,448 / 35,920	May be corrections/refunds; retain for demand
Valid calculated speeds / >80 mph	2,963,754 / 1,107	>80 mph suspicious, not deletion
Duration >180 min / distance >50 mi / fare >$200	1,983 / 412 / 1,070	Upper-tail flags, not automatic invalidity
Pickup / dropoff outside study range	12,018 / 28,083	264/265 are special; see JSON for breakdown

4. Univariate and multivariate evidence

Rules overlap and their counts must not be added. Speed is distance/(duration/60) only for finite nonnegative distance and finite strictly positive duration; otherwise it is missing. The speed-summary denominator is valid calculated speeds, as shown in numeric_summary.csv and the quality table, not all raw rows.

Raw summaries in outputs/numeric_summary.csv show medians and IQRs: distance 1.68 mi (1.00-3.11), duration 11.63 min (7.15-18.68), fare $12.80 (8.60-20.50), passengers 1 (1-1). Extreme raw maxima are flagged; the plotted histograms clip display ranges only.

Sample correlations (50,000 rows, seed 20261108; positive duration, nonnegative distance/fare, speed <=80): fare-distance 0.8136; distance-duration 0.2716. These are associations, not causal effects.

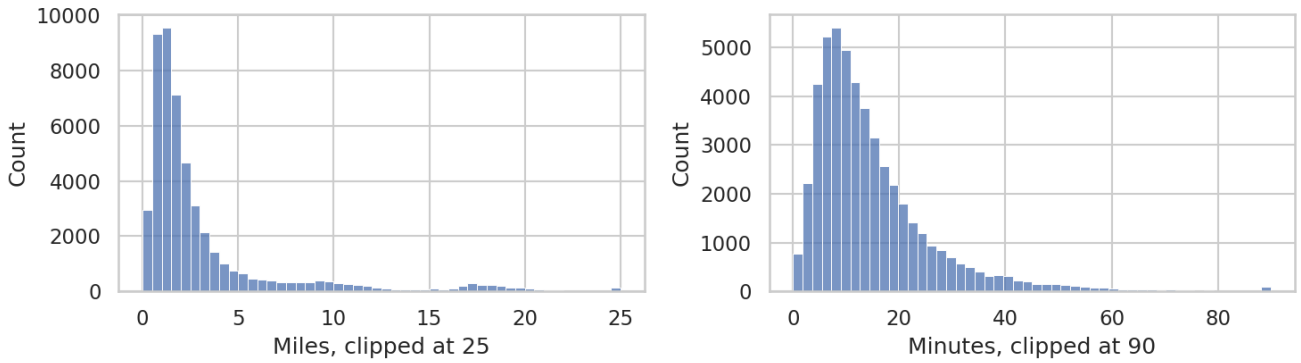


Figure 1. Trip distance and duration on a reproducible diagnostic sample; clipping affects display only.

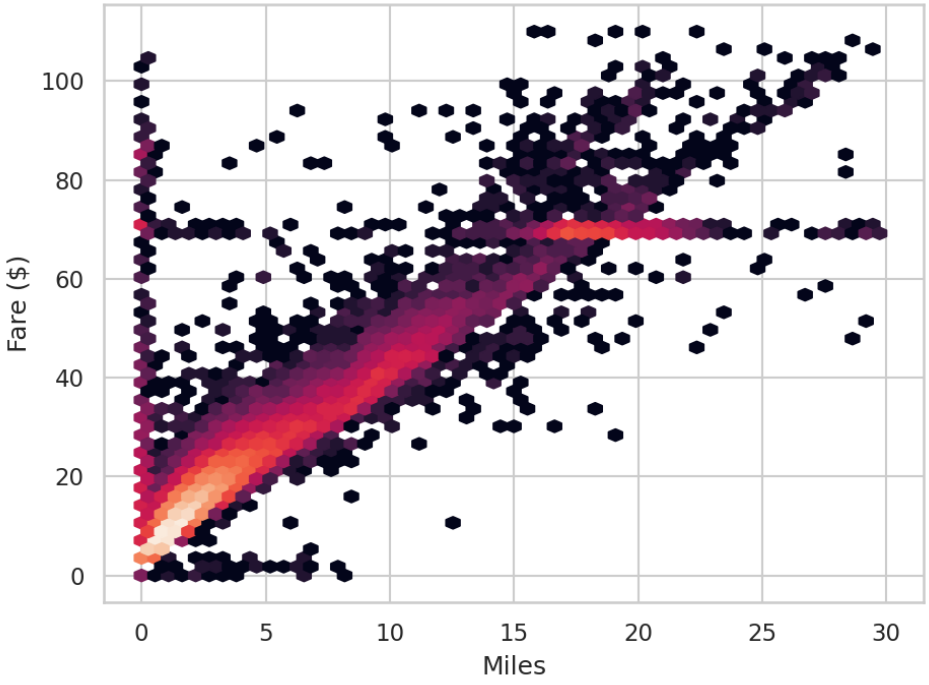


Figure 2. Fare-distance joint distribution, 50,000-row diagnostic sample; plotted ranges are limited.

The hourly comparison, weekday-hour heatmap, fare-distance association and zone concentration describe distinct relationships. Full-population count tables and the 50,000-row trip-variable sample have different denominators; their results must not be mixed.

5. Temporal patterns

Peak pickup hour is 18:00 (212,038 in-month valid-zone submissions). Largest weekday total is Wednesday (493,093), influenced by the number of each weekday in January. Largest date is 2024-01-27 (110,086). See hourly, weekday, daily and weekday-by-hour CSVs. January alone cannot establish annual seasonality.

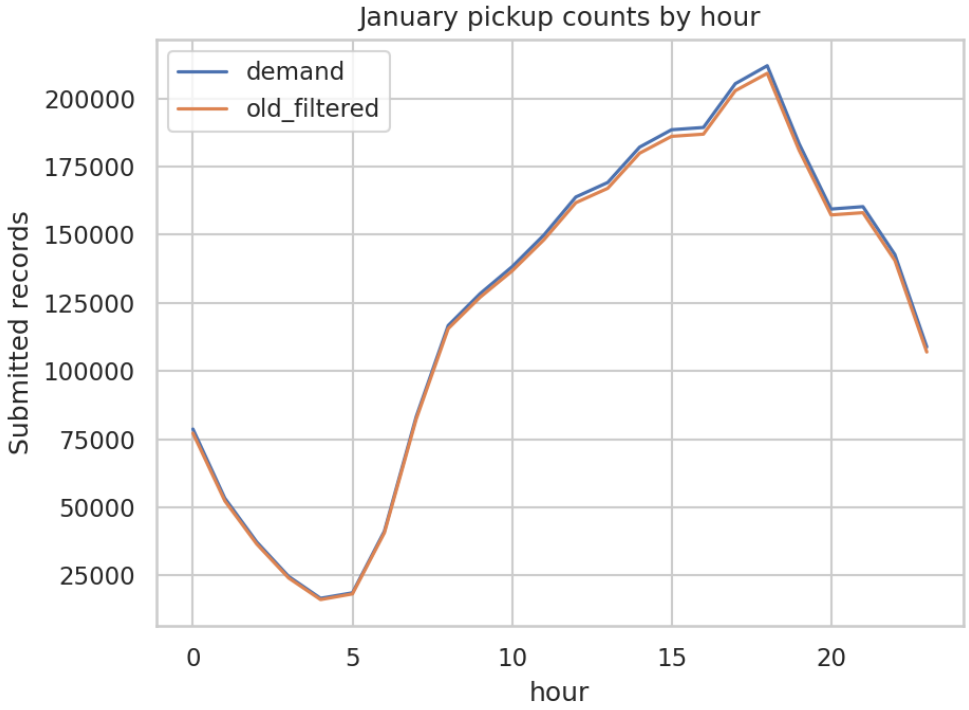


Figure 3. Hourly demand population versus the earlier fare/trip filtered population; the latter undercounts pickup events.

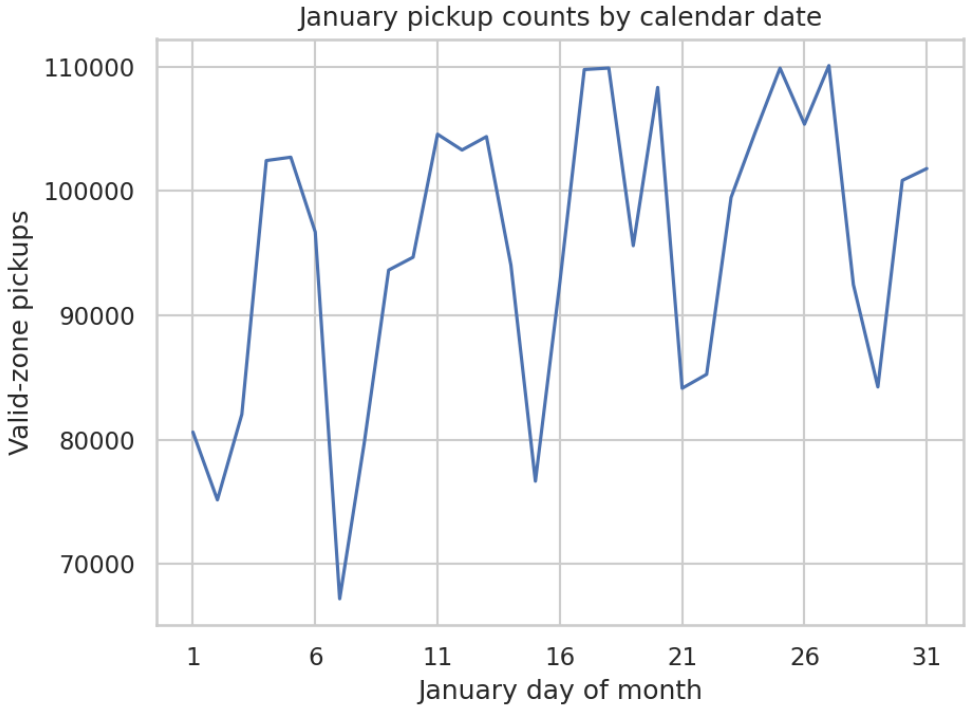


Figure 4. Daily counts for the January valid-zone pickup population.

5. Temporal patterns (continued)

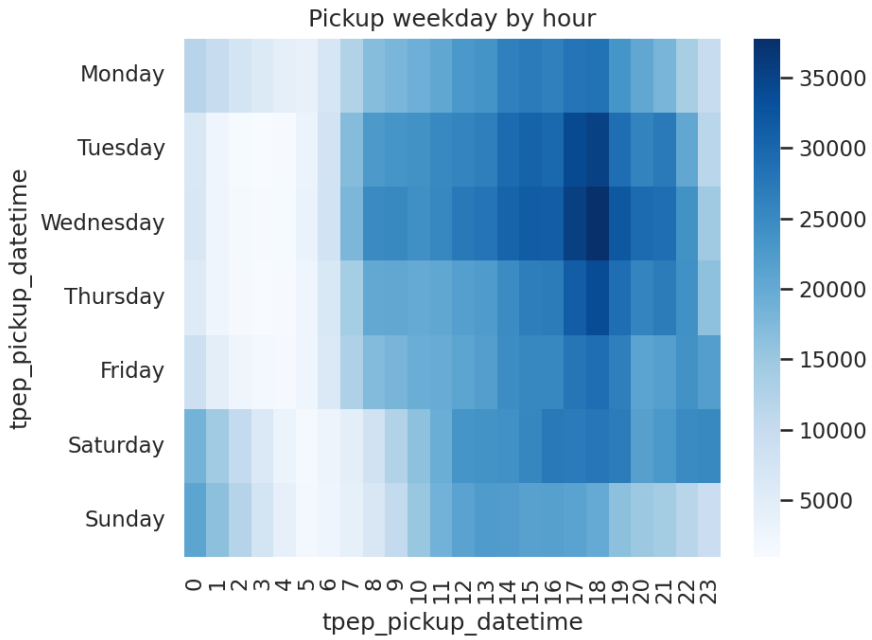


Figure 5. Weekday-hour joint counts; weekday totals are not normalized by number of occurrences.

6. Spatial distribution and zone imbalance

There are 258 observed pickup zones in the demand population and 259 observed dropoff zones among January pickup records with dropoff ID 1-263. These are distinct cohorts: dropoff counts do not require a valid pickup zone. The top 10 pickup zones contribute 39.5425% of accepted-zone January pickups; zone 132 alone has 145,240. Top-10 pickup/dropoff tables contain exact IDs and counts. Missing, zero, official special 264/265 and other out-of-range counts are separated in quality_rules.csv; genuinely unknown lookup keys were not verified. Numeric IDs are not coordinates.

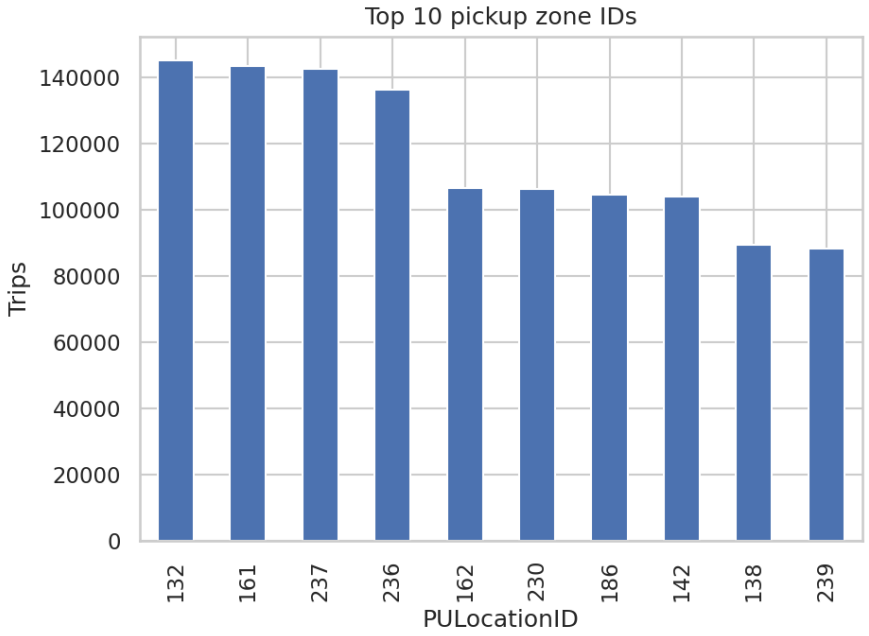


Figure 6. Top 10 pickup zones by count, numeric TLC zone IDs.

7. Research design, limitations and ethics

Three questions:
(1) How do pickup counts differ by zone, hour and day?
(2) Does recent history improve next-hour zone demand forecasting?
(3) Which zone-hours differ unusually from historical patterns?

Q2 is selected for its direct dispatch relevance and testable baseline.

Target and prediction unit. Target for zone z at hour t+1 is the number of pickups within [t+1,t+2); construct a complete zone-hour grid so zero-demand hours appear.

Baseline: past same-zone same-hour-of-week average, with past zone mean fallback.
Metrics. MAE measures average count error; RMSLE limits the dominance of high-volume cells (predictions clipped nonnegative).

Proposed future split: train January-October 2024, validation November, test December; impossible to execute with the current January-only file.

Leakage precautions. Only features known before the forecast origin may enter a model; target-hour counts, dropoff, payment and fare must not enter. Aggregates and transformations must be built on earlier periods.

Future unsupervised work may cluster zone demand profiles; no model is trained in Phase 1.

Aggregate results to avoid exposing fine-grained mobility. Vendor reporting delays, missing fields, coverage restricted to Yellow Taxi, zone-code validity assumptions and observed outliers constrain generalization. No causal claim follows from observational association. Data-use/licensing claims must follow TLC source terms; a separate blanket license is not asserted.

8. Reproducibility

Run scripts/phase1_eda.py on the named Parquet, then scripts/build_phase1_report.py. outputs/audit_summary.json and CSVs are the numerical source for this report. README and outputs/run_log.txt document commands and environment.